Sensor Module



Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium**

27 labs

Covers these lab items: Basic Sensors; Devices such as humidity sensor, temperature sensor, gyro meter, accelerometer, video surveillance cameras; Distance and Gesture Sensor Module; Gas Sensor; Humidity and Temperature Sensor; Humidity Sensor; Inertia Measurement Unit; Inertial Sensors; IR Sensor; LDR Module; Pulse Rate H

What it is used for

A sensor module detects a physical condition such as temperature, humidity, motion, distance, or orientation and converts it into a usable signal. Connect and test the module according to its specifications and the circuit design.

How to use

1. Identify the sensor type, pin labels, operating voltage, and output specifications.
2. Inspect the module and connecting wires for damage or loose pins.
3. Switch off the power before making circuit connections.
4. Connect power, ground, and signal pins according to the datasheet or circuit diagram.
5. Apply the specified power supply and run the approved test program or measurement procedure.
6. Expose the sensor to a suitable test condition and observe the output.
7. Compare readings with expected behaviour and record the results.
8. Switch off power before changing connections and store the module safely.

⚠ Safety

- Never exceed the module's rated voltage or current.
- Check polarity and pinout before connecting power.
- Avoid touching exposed conductors while the circuit is energised.
- Use appropriate ESD precautions for sensitive electronic modules.

Care & storage

- Keep modules dry, clean, and protected from electrostatic discharge.
- Store sensors in suitable packaging and inspect connectors before reuse.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.